



TEST SCENARIOS OF RR VS SRTF

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Scenario A:

Basic mixed workload

- P1: arrival = 0, burst = 8
- P2: arrival = 1, burst = 4
- P3: arrival = 2, burst = 9
- P4: arrival = 3, burst = 5
- Time quantum for RR: 3

Q:

- What is the average waiting time (avg WT)?
- What is the average turnaround time (avg TAT)?
- What is the average response time (avg RT)?
- Which algorithm is more efficient for this scenario?

A:

- Average waiting time for RR: 13.5
- Average waiting time for SRTF: 6.5

(note: SRTF is lower in avg WT)

- Average turnaround time for RR: 20.0
- Average turnaround time for SRTF: 13.0

(note: SRTF is lower in avg TAT)

(result: SRTF is more efficient in terms of WT and TAT)

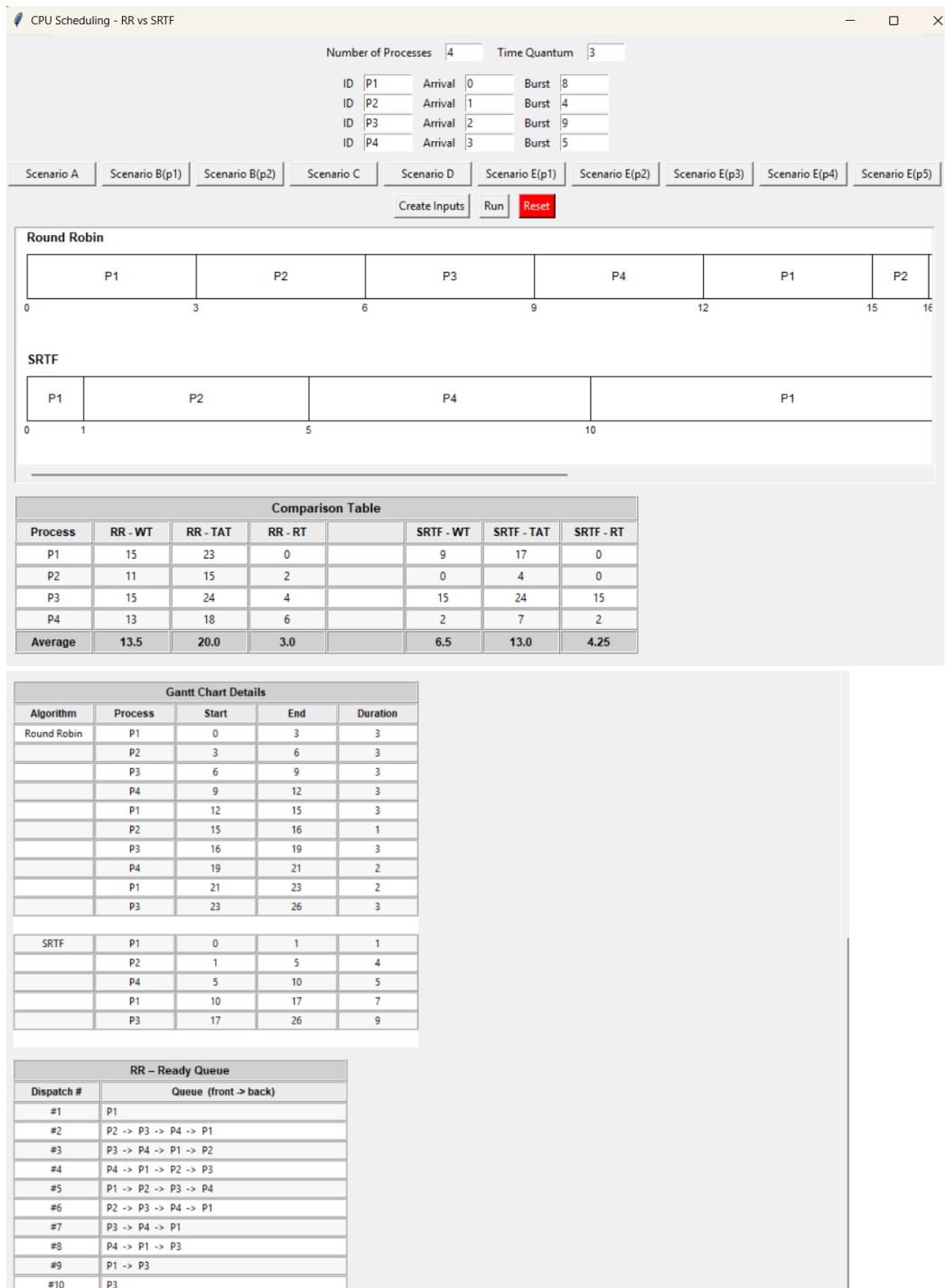
- Average response time for RR: 3.0
- Average response time for SRTF: 4.25

(note: RR is lower in avg RT)

(result: Round Robin gives small advantage in avg RT and fairness of initial CPU access)

- For this scenario, SRTF is better choice if the goal of the whole program is to minimize the avg WT and avg TAT.
While RR is good in avg RT if the program's goal is to prioritize first arrived processes.

Screenshot of Scenario A:



Scenario B:

Quantum sensitivity case

- P1: arrival = 0, burst = 20
- P2: arrival = 0, burst = 3
- P3: arrival = 0, burst = 4
- Time quantum for RR case 1: 2
- Time quantum for RR case 2: 8

Q:

- How does changing time quantum (TQ) affect:
 - Avg WT, avg TAT, avg RT?
- How to compare it to SRTF?

A:

Case 1:

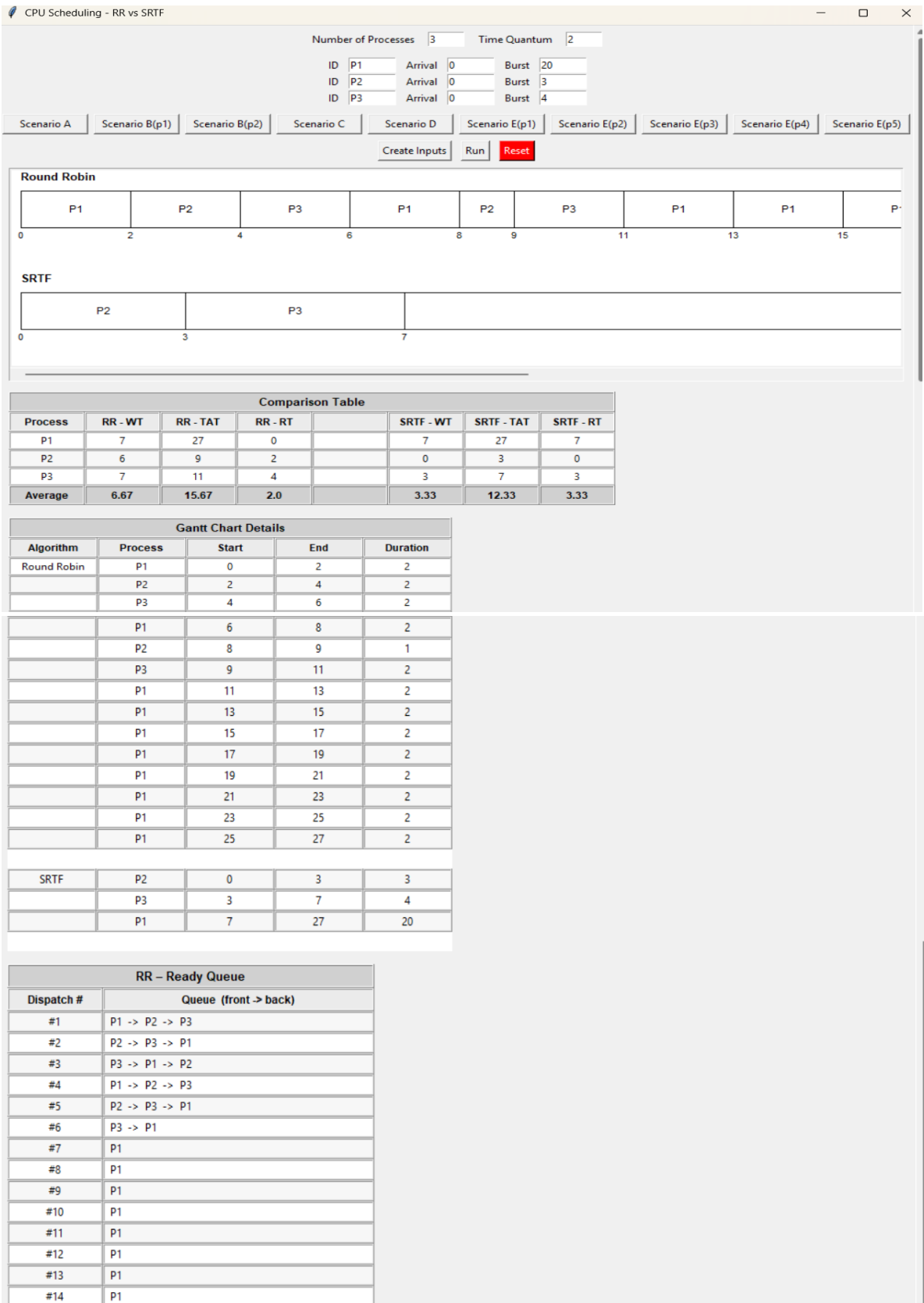
(TQ=2)

- Average waiting time for RR: 6.67
- Average waiting time for SRTF: 3.33
- Average turnaround time for RR: 15.67
- Average turnaround time for SRTF: 12.33
- Average response time for RR: 2.0
- Average response time for SRTF: 3.33

(notes: RR low on avg RT, SRTF low on avg WT and avg TAT)

(results: SRTF is still better than RR in efficiency because it completes short jobs earlier, making it have low avg WT and TAT)

Screenshot of Case 1:



Case 2:

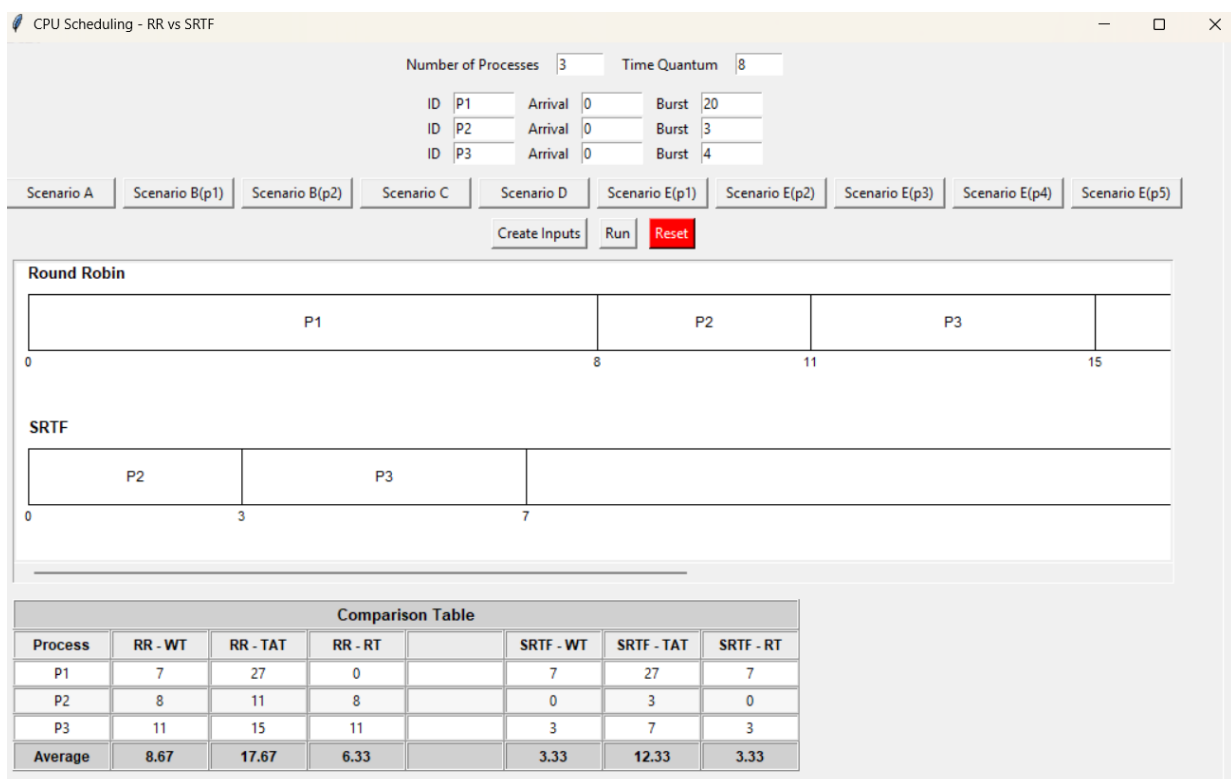
(TQ=8)

- Average waiting time for RR: 8.67
- Average waiting time for SRTF: 3.33
- Average turnaround time for RR: 17.67
- Average turnaround time for SRTF: 12.33
- Average response time for RR: 6.33
- Average response time for SRTF: 3.33

(notes: higher quantum time makes RR behave as FCFS)

(results: SRTF is still better than RR, because SRTF is more stable and strong no matter the quantum chosen for RR)

Screenshot of Case 2:



Gantt Chart Details				
Algorithm	Process	Start	End	Duration
Round Robin	P1	0	8	8
	P2	8	11	3
	P3	11	15	4
	P1	15	23	8
	P1	23	27	4
SRTF	P2	0	3	3
	P3	3	7	4
	P1	7	27	20

RR – Ready Queue	
Dispatch #	Queue (front -> back)
#1	P1 -> P2 -> P3
#2	P2 -> P3 -> P1
#3	P3 -> P1
#4	P1
#5	P1

Scenario C:

Short-job-heavy case

- P1: arrival = 0, burst = 10
- P2: arrival = 1, burst = 1
- P3: arrival = 2, burst = 2
- P4: arrival = 3, burst = 1
- P5: arrival = 4, burst = 2
- Time quantum for RR: 3

Q:

- Does SRTF still give strong advantage for short jobs? (compared to RR)
- How does the average metrics prove that?

A:

- Average waiting time for RR: 3.8
- Average waiting time for SRTF: 1.6
- Average turnaround time for RR: 7.0

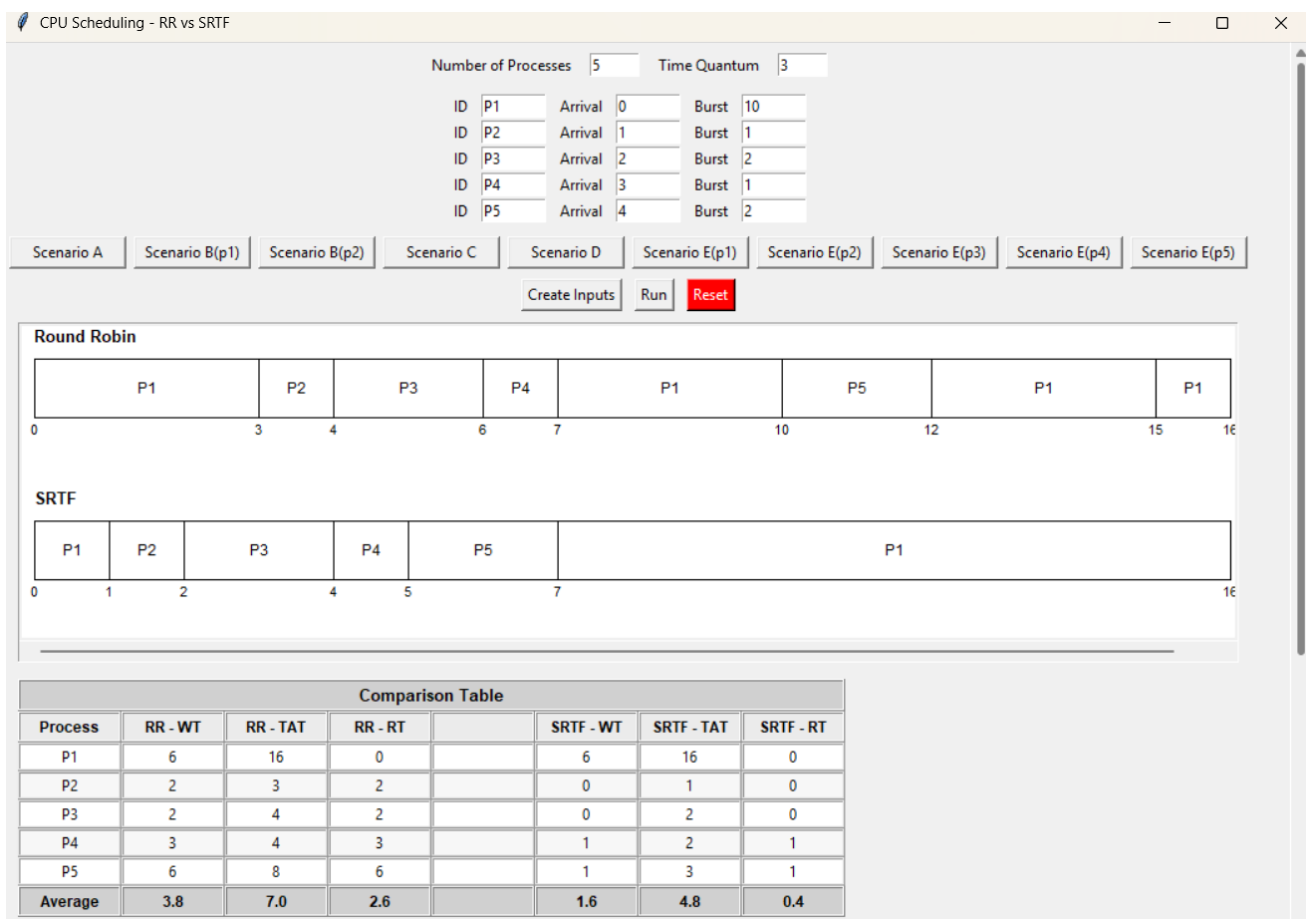
- Average turnaround time for SRTF: 4.8
- Average response time for RR: 2.6
- Average response time for SRTF: 0.4

(notes: SRTF almost gave the immediate CPU access for the short jobs which resulted in low avg WT, TAT, and RT)

(notes: RR allows short jobs to finish quickly but still the processes must wait for their turn in the rotation with P1)

(results: SRTF still gives a strong advantage to short jobs both in completion time and first response. So, SRTF still more efficient than RR for short jobs heavy workloads)

Screenshot for Scenario C:



Gantt Chart Details				
Algorithm	Process	Start	End	Duration
Round Robin	P1	0	3	3
	P2	3	4	1
	P3	4	6	2
	P4	6	7	1
	P1	7	10	3
	P5	10	12	2
	P1	12	15	3
	P1	15	16	1
SRTF	P1	0	1	1
	P2	1	2	1
	P3	2	4	2
	P4	4	5	1
	P5	5	7	2
	P1	7	16	9

RR – Ready Queue	
Dispatch #	Queue (front -> back)
#1	P1
#2	P2 -> P3 -> P4 -> P1
#3	P3 -> P4 -> P1 -> P5
#4	P4 -> P1 -> P5
#5	P1 -> P5
#6	P5 -> P1
#7	P1
#8	P1

Scenario D:

Interactive-style fairness case

- P1: arrival = 0, burst = 20
- P2: arrival = 2, burst = 3
- P3: arrival = 4, burst = 3
- P4: arrival = 6, burst = 3
- Time quantum for RR: 2

Q:

- How do RR and SRTF compare in RT to the interactive processes?

A:

- Average waiting time for RR: 6.5
- Average waiting time for SRTF: 3.0
- Average turnaround time for RR: 13.75
- Average turnaround time for SRTF: 10.25
- Average response time for RR: 1.25
- Average response time for SRTF: 0.75

(notes: both algorithms give interactive processes quick access to the CPU)

(notes: SRTF has lower avg WT, TAT and RT. SRTF has lower RT because short interactive jobs preempt the first process job immediately)

(results: RR provides fairness by rotating the CPU among all processes with small TQ. Yet, SRTF still better in efficiency)

Screenshot for Scenario D:

Number of Processes 4 Time Quantum 2

ID	P1	Arrival	0	Burst	20
ID	P2	Arrival	2	Burst	3
ID	P3	Arrival	4	Burst	3
ID	P4	Arrival	6	Burst	3

Scenario A

Scenario B(p1)

Scenario B(p2)

Scenario C

Scenario D

Scenario E(p1)

Scenario E(p2)

Scenario E(p3)

Scenario E(p4)

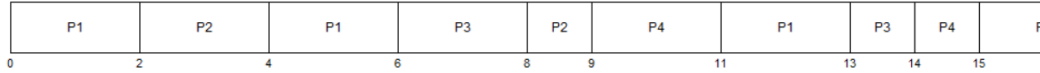
Scenario E(p5)

Create Inputs

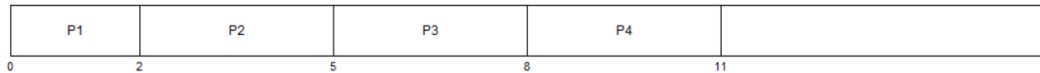
Run

Reset

Round Robin



SRTF



Comparison Table

Process	RR - WT	RR - TAT	RR - RT		SRTF - WT	SRTF - TAT	SRTF - RT
P1	9	29	0		9	29	0
P2	4	7	0		0	3	0
P3	7	10	2		1	4	1
P4	6	9	3		2	5	2
Average	6.5	13.75	1.25		3.0	10.25	0.75

Gantt Chart Details

Algorithm	Process	Start	End	Duration
Round Robin	P1	0	2	2
	P2	2	4	2
	P1	4	6	2
	P3	6	8	2
	P2	8	9	1
	P4	9	11	2
	P1	11	13	2
	P3	13	14	1
	P4	14	15	1
	P1	15	17	2
	P1	17	19	2
	P1	19	21	2
	P1	21	23	2
	P1	23	25	2
	P1	25	27	2
	P1	27	29	2
SRTF	P1	0	2	2
	P2	2	5	3
	P3	5	8	3
	P4	8	11	3
	P1	11	29	18

RR – Ready Queue	
Dispatch #	Queue (front -> back)
#1	P1
#2	P2 -> P1
#3	P1 -> P3 -> P2
#4	P3 -> P2 -> P4 -> P1
#5	P2 -> P4 -> P1 -> P3
#6	P4 -> P1 -> P3
#7	P1 -> P3 -> P4
#8	P3 -> P4 -> P1
#9	P4 -> P1
#10	P1
#11	P1
#12	P1
#13	P1
#14	P1
#15	P1
#16	P1

Scenario E:

Validation case

Q:

- How does the system respond to invalid process data?
- How is it shown in the GUI?

Case 1:

- P1: arrival = 0, burst = 5
- P2: arrival = 1, burst = 3
- Time quantum for RR: 2

(results: runs normally)

Screenshot of Case 1:

CPU Scheduling - RR vs SRTF

Number of Processes: 2 Time Quantum: 2

ID P1 Arrival 0 Burst 5
ID P2 Arrival 1 Burst 3

Scenario A Scenario B(p1) Scenario B(p2) Scenario C Scenario D Scenario E(p1) Scenario E(p2) Scenario E(p3) Scenario E(p4) Scenario E(p5)

Create Inputs Run Reset

Round Robin

0 2 4 6 7 8

P1 P2 P1 P2 P1

SRTF

0 1 4 8

P1 P2 P1

Comparison Table

Process	RR - WT	RR - TAT	RR - RT		SRTF - WT	SRTF - TAT	SRTF - RT
P1	3	8	0		3	8	0
P2	3	6	1		0	3	0
Average	3.0	7.0	0.5		1.5	5.5	0.0

Gantt Chart Details

Algorithm	Process	Start	End	Duration
Round Robin	P1	0	2	2
	P2	2	4	2
	P1	4	6	2
	P2	6	7	1
	P1	7	8	1
SRTF	P1	0	1	1
	P2	1	4	3
	P1	4	8	4

RR - Ready Queue

Dispatch #	Queue (front -> back)
#1	P1
#2	P2 -> P1
#3	P1 -> P2
#4	P2 -> P1
#5	P1

Test cases of invalid inputs:

Case 2:

- P1: arrival = -1, burst = 5
- P2: arrival = 1, burst = 3
- Time quantum for RR: 2

(results: shows an error “Arrival time of P1 cannot be negative”)

Screenshot of Case 2:

The screenshot shows a web application titled "CPU Scheduling - RR vs SRTF". At the top, there are input fields for "Number of Processes" (set to 2) and "Time Quantum" (set to 2). Below these are input fields for process details:

ID	Arrival	Burst
P1	-1	5
P2	1	3

Below the input fields are buttons for "Scenario A", "Scenario B(p1)", "Scenario B(p2)", "Scenario C", "Scenario D", "Scenario E(p1)", "Scenario E(p2)", "Scenario E(p3)", "Scenario E(p4)", and "Scenario E(p5)". There are also buttons for "Create Inputs", "Run", and "Reset".

The main area displays two Gantt charts:

Round Robin

Process	Start	End
P1	0	2
P2	2	4
P1	4	6
P2	6	7
P1	7	8

SRTF

Process	Start	End
P1	0	1
P2	1	4
P1	4	5

An error dialog box is displayed over the SRTF chart with the message: "Error: Arrival time of P1 cannot be negative".

Below the Gantt charts is a "Comparison Table":

Process	RR - WT	RR - TAT	RR - RT	SRTF - WT	SRTF - TAT	SRTF - RT
P1	3	8	0	3	8	0
P2	3	6	1	0	3	0
Average	3.0	7.0	0.5	1.5	5.5	0.0

At the bottom is a "Gantt Chart Details" table:

Algorithm	Process	Start	End	Duration
Round Robin	P1	0	2	2
	P2	2	4	2
	P1	4	6	2
	P2	6	7	1
	P1	7	8	1

Case 3:

- P1: arrival = 0, burst = -3
- P2: arrival = 1, burst = 0
- Time quantum for RR: 2

(results: shows an error “Burst time of P1 must be positive integer”)

Screenshot of Case 3:

The screenshot shows a CPU Scheduling simulation interface titled "CPU Scheduling - RR vs SRTF". The interface includes input fields for "Number of Processes" (2) and "Time Quantum" (2). Below these are input fields for process details:

ID	Arrival	Burst
P1	0	-3
P2	1	0

Below the input fields are buttons for "Scenario A", "Scenario B(p1)", "Scenario B(p2)", "Scenario C", "Scenario D", "Scenario E(p1)", "Scenario E(p2)", "Scenario E(p3)", "Scenario E(p4)", and "Scenario E(p5)". There are also buttons for "Create Inputs", "Run", and "Reset".

The "Round Robin" scheduling algorithm is selected, and its Gantt chart is displayed:

Process	Start	End	Duration
P1	0	2	2
P2	2	4	2
P1	4	6	2
P2	6	7	1
P1	7	8	1

The "SRTF" scheduling algorithm is also shown with its Gantt chart:

Process	Start	End	Duration
P1	0	1	1
P2	1	4	3
P1	4	5	1

An error message dialog box is displayed over the SRTF Gantt chart, stating: "Error: Burst time of P1 must be a positive integer".

Below the Gantt charts is a "Comparison Table":

Process	RR - WT	RR - TAT	RR - RT	SRTF - WT	SRTF - TAT	SRTF - RT
P1	3	8	0	3	8	0
P2	3	6	1	0	3	0
Average	3.0	7.0	0.5	1.5	5.5	0.0

Below the comparison table is a "Gantt Chart Details" table:

Algorithm	Process	Start	End	Duration
Round Robin	P1	0	2	2
	P2	2	4	2
	P1	4	6	2
	P2	6	7	1
	P1	7	8	1

Case 4:

- P1: arrival = 0, burst = 5
- P1: arrival = 1, burst = 3
- Time quantum for RR: 2

(results: shows an error “Duplicate ID:1”)

Screenshot of Case 4:

CPU Scheduling - RR vs SRTF

Number of Processes: 2 Time Quantum: 2

ID: P1 Arrival: 0 Burst: 5
ID: P1 Arrival: 1 Burst: 3

Scenario A Scenario B(p1) Scenario B(p2) Scenario C Scenario D Scenario E(p1) Scenario E(p2) Scenario E(p3) Scenario E(p4) Scenario E(p5)

Create Inputs Run Reset

Round Robin

P1	P2	P1	P2	P1
0	2	4	6	7

SRTF

P1	P2	P1
0	1	4

Error
Duplicate ID: P1
OK

Comparison Table

Process	RR - WT	RR - TAT	RR - RT		SRTF - WT	SRTF - TAT	SRTF - RT
P1	3	8	0		3	8	0
P2	3	6	1		0	3	0
Average	3.0	7.0	0.5		1.5	5.5	0.0

Case 5:

- P1: arrival = 0, burst = 5
- P2: arrival = 1, burst = 3
- Time quantum for RR: 0

(results: shows an error “invalid quantum time quantum must be greater than 0”)

Screenshot of Case 5:

The screenshot shows a software application titled "CPU Scheduling - RR vs SRTF". At the top, there are input fields for "Number of Processes" (set to 2) and "Time Quantum" (set to 0). Below these are input fields for two processes, P1 and P2, with their respective arrival and burst times. P1 has an arrival time of 0 and a burst time of 5. P2 has an arrival time of 1 and a burst time of 3. There are several scenario buttons: "Scenario A", "Scenario B(p1)", "Scenario B(p2)", "Scenario C", "Scenario D", "Scenario E(p1)", "Scenario E(p2)", "Scenario E(p3)", "Scenario E(p4)", and "Scenario E(p5)". Below the buttons are "Create Inputs", "Run", and "Reset" buttons. The "Run" button is highlighted in red.

Below the buttons, there are two Gantt charts. The first is for "Round Robin" scheduling, showing a sequence of P1, P2, P1, P2, P1 over a timeline from 0 to 8. The second is for "SRTF" scheduling, showing a sequence of P1, P2, P1 over a timeline from 0 to 4.

An error dialog box is displayed in the foreground, titled "Error", with a red 'X' icon. The message inside the dialog is "Invalid Quantum: Time Quantum must be greater than 0". There is an "OK" button at the bottom right of the dialog.

Below the Gantt charts, there is a "Comparison Table" and a "Gantt Chart Details" table.

Process	RR - WT	RR - TAT	RR - RT		SRTF - WT	SRTF - TAT	SRTF - RT
P1	3	8	0		3	8	0
P2	3	6	1		0	3	0
Average	3.0	7.0	0.5		1.5	5.5	0.0

Algorithm	Process	Start	End	Duration
Round Robin	P1	0	2	2
	P2	2	4	2
	P1	4	6	2
	P2	6	7	1
	P1	7	8	1